MULTI-VM DEPARTMENTAL

HOMELAB BUILD

System Administrator Build Documentation — 5 Departments • 15 Virtual Machines • Virtualised Network Segmentation

Platform: Oracle VirtualBox | Servers: Windows Server 2022 | Workstations: Windows 8 | Hypervisor: Type 2

15	5	5	10	1
Virtual Machines	Departments	Servers (WS2022)	PCs (Windows 8)	Day Build Time

Date: 7TH AUGUST 2025

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1. Executive Summary

This project is an authoritative build record for a multi-virtual-machine departmental homelab constructed on Oracle VirtualBox. The lab simulates a segmented enterprise network topology across five organizational departments: Finance, Audit, Information Technology, Procurement, and Guest. It was designed and built by a system administrator to replicate the structure, separation, and management characteristics of a production corporate network in a fully virtualized, consequence-free environment.

The lab consists of fifteen virtual machines assets of which five departmental servers running Windows Server 2022, ten workstations running Windows 8, including a Guest segment with two workstations and separate server. Each department's virtual machines are logically grouped within VirtualBox using the Groups feature, which provides a visual and administrative structure mirroring real-world organizational boundaries. Network isolation is enforced using VirtualBox's Internal Network feature, ensuring that each department's traffic remains within its own broadcast domain.

This homelab serves as a practical platform for system administration skill development and demonstration — providing a hands-on environment for practicing Active Directory, Group Policy, user management, network configuration, incident simulation, and security hardening without risk to production systems. Every configuration decision documented here reflects the standards and decision-making processes of a professional system administrator.

2. Lab Objectives & Sysadmin Rationale

2.1 Why Build a Departmental Multi-VM Lab?

A single-machine lab teaches OS installation. A departmental multi-VM lab teaches how systems relate to one another — which is the work of a system administrator. By building separate machines for separate departments, this lab enables practice of the scenarios that define real-world sysadmin work:

- User account management across a multi-department organisation
- Applying different security policies (Group Policy Objects) per department
- Simulating network segmentation and understanding why Sales should not be able to reach Guest network resources
- Server role management — each departmental server is a candidate for DHCP, DNS, file sharing, or domain controller promotion
- Troubleshooting cross-department connectivity and access issues
- Incident response simulation — isolating a compromised department VM without affecting others

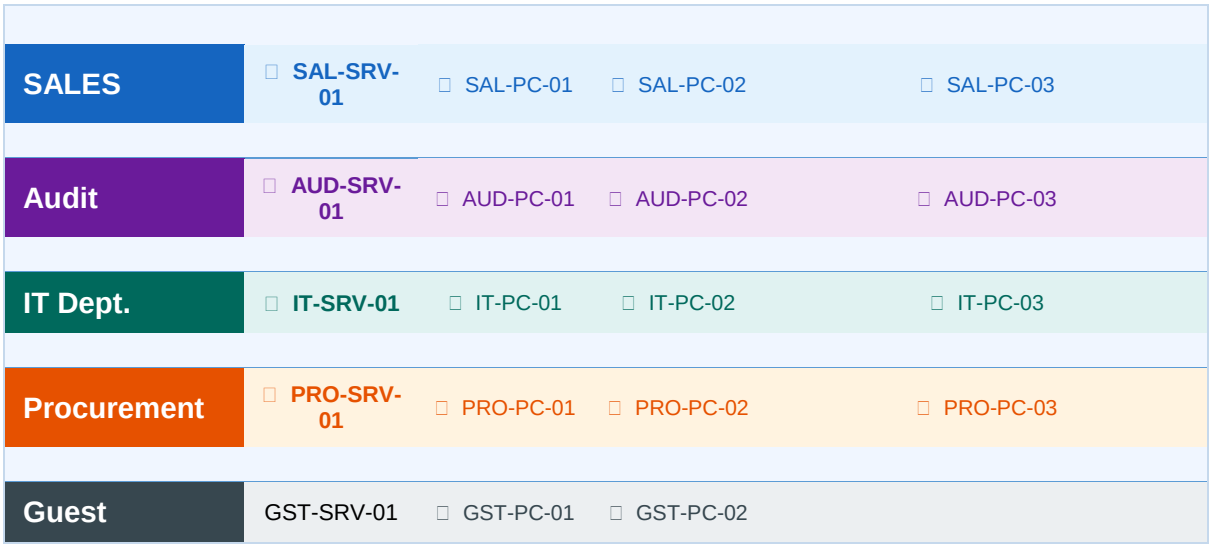
2.2 Design Principles

Design Principle	How This Lab Applies It
Separation of Concerns	Each department is an isolated logical unit — separate server, separate workstations, separate internal network. No department shares resources with another by default.
Least-Privilege by Default	Guest department has a completely separated server isolated from main network function — simulating the real-world restriction that guest/visitor devices should have minimal access and no domain resources.
Realistic Naming Conventions	All VMs follow a structured hostname convention (e.g. SAL-SRV-01, AUD-PC-02) matching enterprise standards — making the lab documentation transferable to real environments.
Scalability	The VirtualBox Groups structure and network design allow new departments or VMs to be added without redesigning the existing topology.
Documentation as a Deliverable	This build document is itself a sysadmin deliverable — demonstrating that configuration decisions were intentional, recorded, and reproducible.

3. Network Topology Design (Reference Diagram)

Figure 1 below is the reference network topology diagram for this lab. It shows the five departmental segments, each represented as a horizontal band with its server and workstations. The blue horizontal lines represent the network boundary between each department — in VirtualBox terms, these are separate Internal Networks, ensuring that VMs in one department cannot communicate directly with VMs in another without routing through a dedicated gateway (a future extension to this lab).

Figure 1 — Departmental Homelab Network Topology (Reference Design)



Note: Each horizontal blue line in the reference image (see attached Figure 1 — wwww.PNG) represents a separate internal network boundary in VirtualBox. This document recreates that topology programmatically above and uses it as the authoritative design reference for all VM and network configurations that follow.

4. IP Addressing & VM Naming Convention

4.1 Hostname Naming Standard

A structured hostname convention was defined before any VM was created. Consistent naming is a fundamental sysadmin discipline — it enables rapid identification of any machine's department and role from its hostname alone, and aligns with real-world enterprise naming standards.

Hostname Convention: [DEPT-CODE]-[ROLE]-[INDEX]

Format: [3-letter Department Code] - [Role: SRV = Server | PC = Workstation] - [2-digit index]

Examples:

- SAL-SRV-01 = SALES Department, Server 1
- AUD-PC-02 = Audit Department, Workstation 2
- IT-SRV-01 = IT Department, Server 1
- GST-PC-02 = Guest Department, Workstation 2

This convention is consistent with Microsoft naming guidance for Active Directory environments and ensures that VirtualBox Groups, DNS records (if configured), and event logs all use identical identifiers.

4.2 IP Addressing Scheme

We are using a separate /27 subnets which divide 192.168.0.0/24 into blocks of 32 addresses which were allocated to each department in block. This mirrors real-world VLAN design where each department operates in its own broadcast domain with a dedicated subnet, enabling per-department firewall rules and access control to be applied in future lab extensions.

Department	Subnet	Subnet Mask	Server IP	PC IP Range
SALE	192.168.0.0/27	255.255.255.224	DHCP	192.168.0.1 – .30
Audit	192.168.0.32/27	255.255.255.224	DHCP	192.168.0.33 – .62
IT Dept.	192.168.0.64/27	255.255.255.224	DHCP	192.168.0.65 – .94
Procurement	192.168.0.96/27	255.255.255.224	DHCP	192.168.0.97 – .126
Guest	192.168.0.128/27	255.255.255.224	DHCP	192.168.0.129 – .158

5. Host Machine Requirements & Pre-Build Checklist

Running 18 concurrent virtual machines is resource-intensive. The following specifications and pre-build checks were verified before any VM creation began. A sysadmin never starts a multi-VM build without validating the host can support the intended load.

Component	Minimum Required	Recommended / Used
CPU	Intel/AMD with VT-x / AMD-V enabled in BIOS (required for 64-bit guest OSes)	8-core or higher — essential for running 18 VMs without critical performance degradation
RAM	32 GB (absolute minimum for 18 VMs)	64 GB — allows 2–4 GB per VM with headroom for the host OS
Storage	500 GB free disk space (dynamically allocated VMs will grow over time)	1 TB SSD — VMs on SSD vs HDD reduces boot times by ~70% and improves responsiveness
OS	Windows 10/11 or Linux (Ubuntu) as host OS	Windows 11 Pro used as host in this build
VirtualBox	VirtualBox 7.x (latest stable release)	VirtualBox 7.x downloaded from virtualbox.org
Network	Any standard NIC (VirtualBox handles virtualised networking internally)	Single NIC — VirtualBox internal networks are host-independent

5.1 RAM Allocation Plan (15-18 VMs)

RAM Allocation Strategy — Avoid Host Starvation

Total lab RAM budget: ~24 GB (leaving ~8 GB for host OS on a 32 GB machine)

- SAL-SRV-01: 4 GB (Windows Server 2022 needs minimum 2 GB; 4 GB recommended)
- AUD-SRV-01: 4 GB
- IT-SRV-01: 4 GB
- PRO-SRV-01: 4 GB
- 14 × Win8 PCs: 1 GB each = 14 GB
- Total: 30 GB (not all VMs run simultaneously — run 1–2 departments at a time)

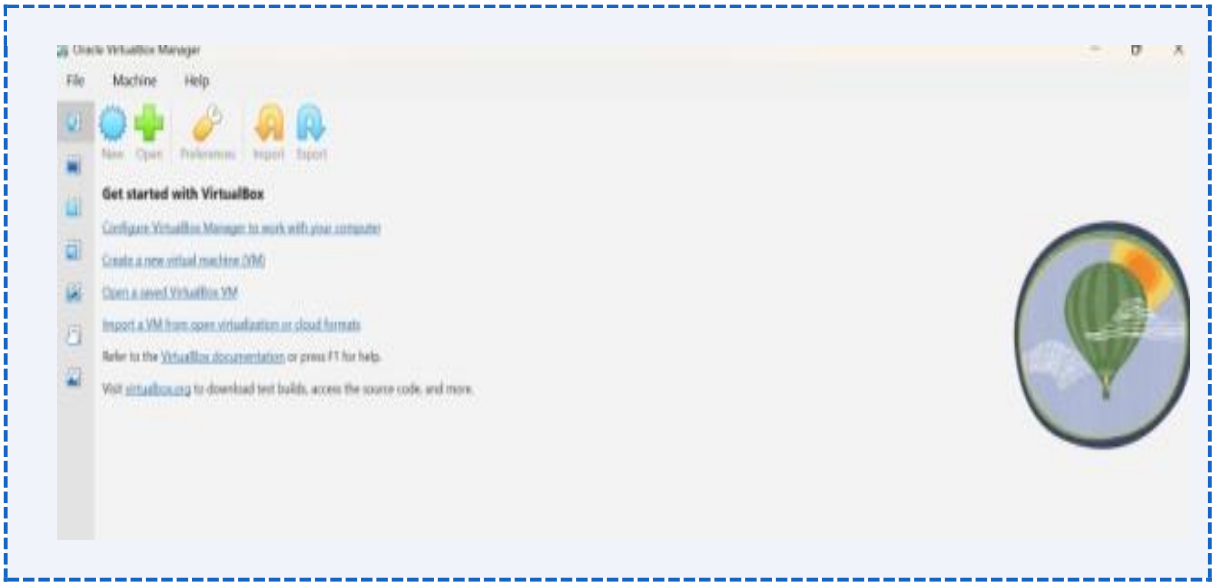
Sysadmin Note: Running all 18 VMs simultaneously is only feasible on 64 GB RAM. For a 32 GB host, operate one department group at a time. VirtualBox's Groups feature makes this straightforward — power on/off entire groups with one click.

6. Software Downloads & Installation

6.1 Oracle VirtualBox Installation

VirtualBox is the Type 2 hypervisor (hosted hypervisor) that provides the virtualization layer for this entire lab. It was downloaded from the official Oracle website to ensure authenticity and to receive the latest version with current guest OS support.

1. Navigated to https://www.virtualbox.org/wiki/Downloads
2. Selected the platform package for the host OS (Windows hosts for a Windows 11 host machine)
3. Downloaded the VirtualBox installer (VirtualBox-[version]-Win.exe)
4. Also downloaded the VirtualBox Extension Pack from the same page — provides USB 2.0/3.0, RDP, and disk encryption support
5. Ran the VirtualBox installer as Administrator — accepted all default settings
6. After VirtualBox installation completed, ran the Extension Pack installer — accepted the Oracle VirtualBox Extension Pack License Agreement
7. Launched VirtualBox — confirmed the Manager window opened with no VMs listed (clean state)



6.2 Windows Server 2022 ISO

Windows Server 2022 provides the operating system for the four departmental servers. The evaluation ISO was obtained directly from Microsoft — no licence key is required for the 180-day evaluation period, which is more than sufficient for a homelab environment.

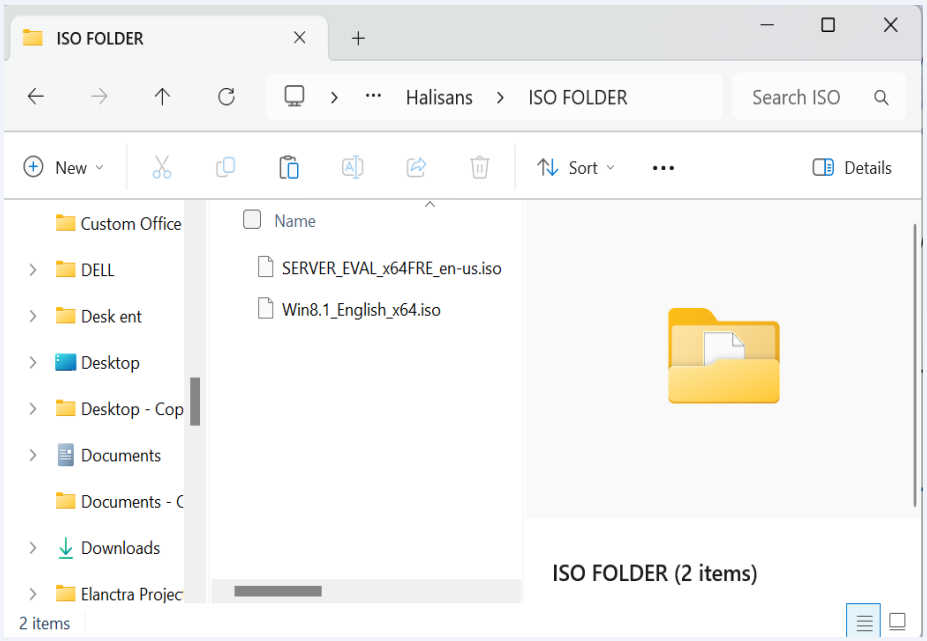
8. Navigated to https://www.microsoft.com/en-us/evalcenter/download-windows-server-2022
9. Selected ISO download, filled in the registration form (name, email, company)

10. Selected English (United States) and 64-bit edition
11. Downloaded the ISO file: SERVER_EVAL_x64FRE_en-us_OEM.iso (approximately 5.0 GB)
12. Verified the file downloaded completely by checking the file size matched the listed download size

6.3 Windows 8 ISO

Windows 8 was selected for the workstation VMs. As a lighter-weight OS compared to Windows 10/11, it reduces RAM and disk consumption per VM — important when running 14 workstations. The ISO was sourced from a legitimate Microsoft archive.

13. Sourced the Windows 8 x64 ISO from the Microsoft Software Download centre or a legitimate ISO archive
14. Downloaded the ISO file: Win8_[edition]_English_x64.iso (approximately 3.0–3.5 GB)
15. Stored both ISOs in a dedicated folder: C:\VM-ISOs\ (or equivalent) for easy access during VM creation



ISO Files — Both ISOs in the VM-ISOs Folder

Capture Windows Explorer (or File Manager) showing the VM-ISOs folder containing both ISO files: the Windows Server 2022 ISO and the Windows 8 ISO, with their file sizes visible confirming successful download.

7. VM Specifications — Full Inventory

The following table is the complete VM inventory for the homelab. These specifications were determined before any VM was created and serve as the build reference throughout Sections 8–9. All VMs use dynamically allocated storage to conserve host disk space.

Hostname	Role	OS	RAM	CPU	Disk	VirtualBox Group
SAL-SRV-01	Dept. Server	Windows Server 2022	4 GB	4	50 GB	Sales/Sale-Server
SAL-PC-01	Workstation	Windows 8	2 GB	2	40 GB	Sale/Sale-PCs
SAL-PC-02	Workstation	Windows 8	2GB	2	25 GB	Sale/Sale-PCs
SAL-PC-03	Workstation	Windows 8	2 GB	2	25 GB	Sale/Sale-PCs
AUD-SRV-01	Dept. Server	Windows Server 2022	4 GB	2	60 GB	Audit/Audit-Server
AUD-PC-01	Workstation	Windows 8	1 GB	1	25 GB	Audit/Audit-PCs
AUD-PC-02	Workstation	Windows 8	1 GB	1	25 GB	Audit/Audit-PCs
AUD-PC-03	Workstation	Windows 8	1 GB	1	25 GB	Audit/Audit-PCs
IT-SRV-01	Dept. Server	Windows Server 2022	4 GB	2	60 GB	IT-Dept/IT-Server
IT-PC-01	Workstation	Windows 8	1 GB	1	25 GB	IT-Dept/IT-PCs
IT-PC-02	Workstation	Windows 8	1 GB	1	25 GB	IT-Dept/IT-PCs
IT-PC-03	Workstation	Windows 8	1 GB	1	25 GB	IT-Dept/IT-PCs
PRO-SRV-01	Dept. Server	Windows Server 2022	4 GB	2	60 GB	Procurement/PRO-Server
PRO-PC-01	Workstation	Windows 8	1 GB	1	25 GB	Procurement/PRO-PCs
PRO-PC-02	Workstation	Windows 8	1 GB	1	25 GB	Procurement/PRO-PCs
PRO-PC-03	Workstation	Windows 8	2 GB	2	25 GB	Procurement/PRO-PCs
GST-PC-01	Workstation	Windows 8	2 GB	2	25 GB	Guest/Guest-PCs

GST-PC-02	Workstation	Windows 8	2 GB	2	25 GB	Guest/Guest-PCs

8. Building the Virtual Machines

All VMs were created using the VirtualBox New VM Wizard. The procedure below is documented once in full for the Sales department (as the template), with abbreviated records for the remaining departments noting only the values that differ. This mirrors real sysadmin documentation practice — establish the standard once, then record deviations.

• **SALES Department**	**3 VMs VMs total**	1 Server	2 Workstations

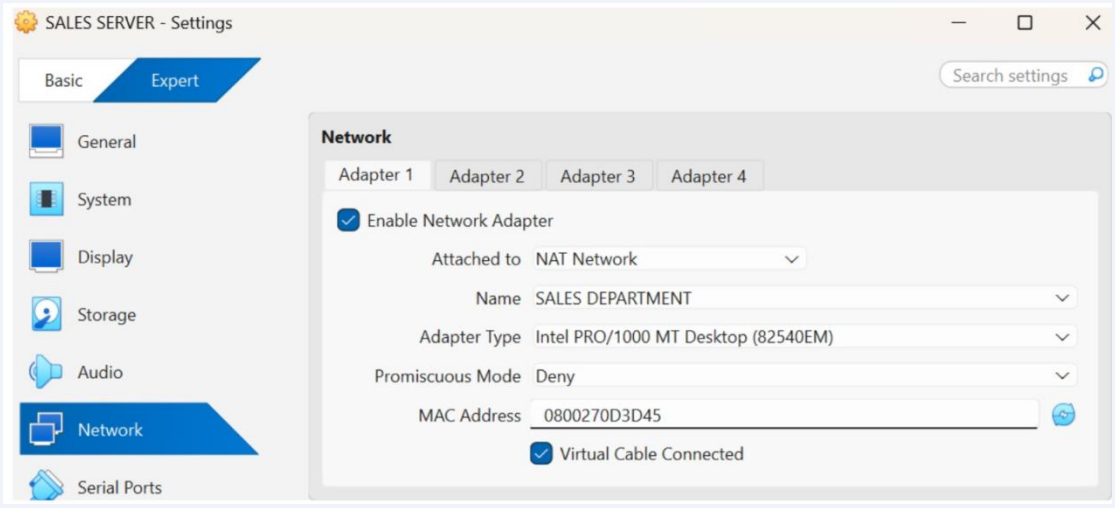
8.1 Sales Department VMs

SALES-SERVER — SALES Department Server

16. In VirtualBox Manager, clicked Machine > New (or Ctrl+N)
17. Name: SALE SERVER | Folder: [choose a dedicated VM storage folder e.g. D:\VMs\SALES\]
18. ISO Image: Selected the Windows Server 2022 ISO from C:\VM-ISOs\
19. Edition: Windows Server 2022 Standard (Desktop Experience) — checked 'Skip Unattended Installation'
20. Hardware: RAM = 4096 MB | Processors = 4 CPUs
21. Hard Disk: Create a Virtual Hard Disk Now | Size = 50 GB | VDI | Dynamically Allocated
22. Clicked Finish — VM appeared in the VirtualBox Manager list
23. Right-clicked SALE SERVERSALE1 > Settings > System > Boot Order: set Optical first, then Hard Disk
24. Settings > Display > Video Memory: set to 128 MB
25. Settings > Network > Adapter 1: Attached to = NAT Network | Name = SALE-Net



VirtualBox — SAL-SRV-01 VM Summary Screen (After Creation)

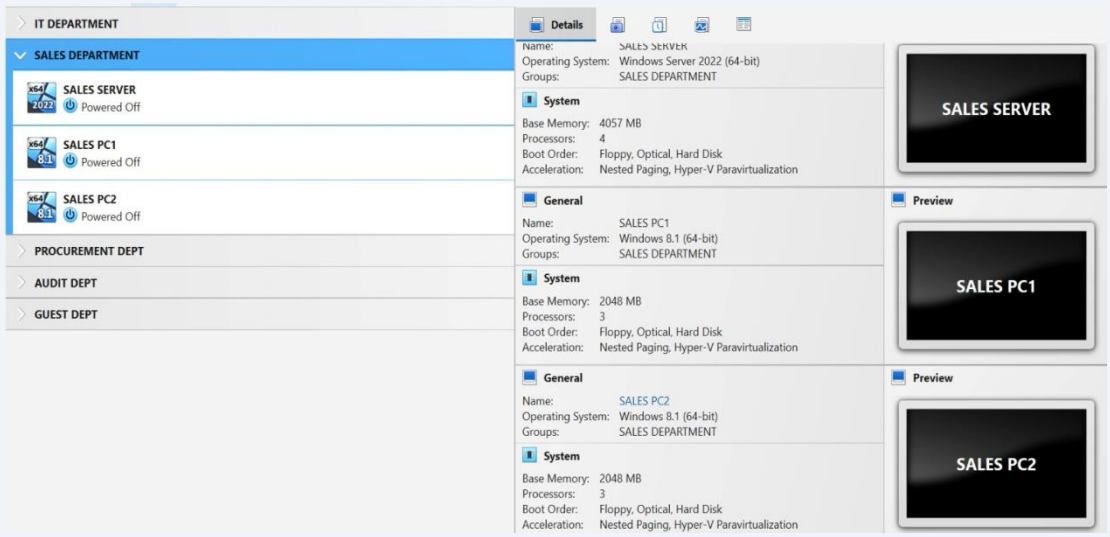


VirtualBox — SAL-SRV-01 Settings: Network Tab

SALE-PC-01, SALE-PC-02, — Sales Workstations

The two sales workstations were created using the same procedure as SALE-SRV-01 with the following differences:

Setting	SAL-PC-01	SALE-PC-02	SALE-PC-03
Name	SAL-PC-01	SAL-PC-02	SAL-PC-03
ISO	Windows 8 x64	Windows 8 x64	Windows 8 x64
RAM	4096 MB	4096 MB	4096 MB
vCPUs	4	4	4
Disk	40 GB VDI Dyn.	40 GB VDI Dyn.	40 GB VDI Dyn.
Network	`Sales-Net`	`Sales-Net`	`Sales-Net`

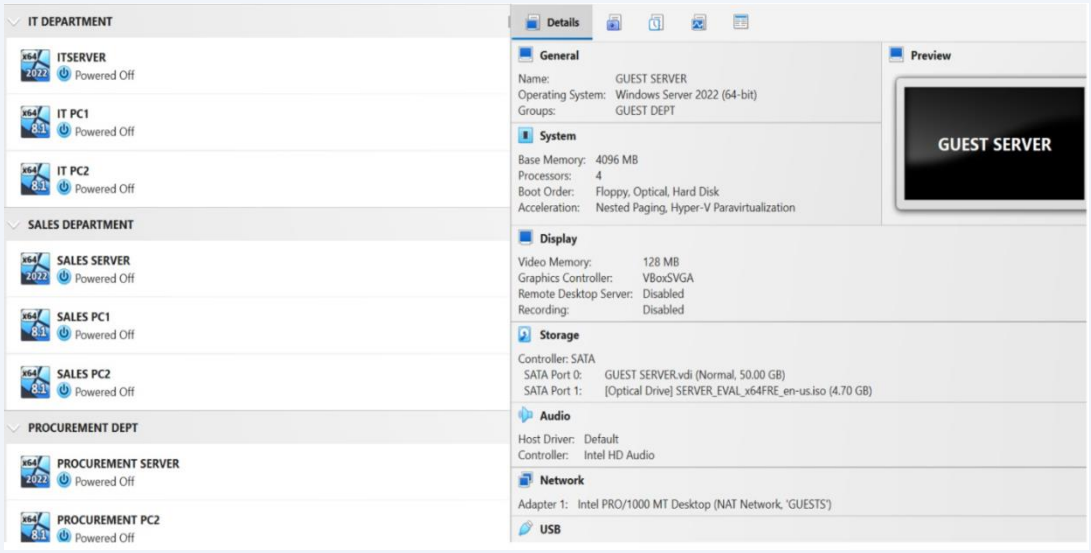


VirtualBox — All Sales PCs Listed (SAL-PC-01, 02

8.2 Audit Department VMs

All three Audit VMs were created following the identical procedure established in Section 8.1, substituting Audit-specific values. The key distinction is the Internal Network name — all Audit VMs connect to Audit-Net, not sale-Net, ensuring complete isolation between the two departments.

Setting	AUD-SRV-01	AUD-PC-01	AUD-PC-02	AUD-PC-03
OS	Windows Server 2022	Windows 8	Windows 8	Windows 8
RAM	4096 MB	4096 MB	4096 MB	4096 MB
vCPUs	4	4	4	4
Disk	50 GB VDI Dyn.	40 GB VDI	40 GB VDI	40 GB VDI
Network	Audit-Net	Audit-Net	Audit-Net	Audit-Net



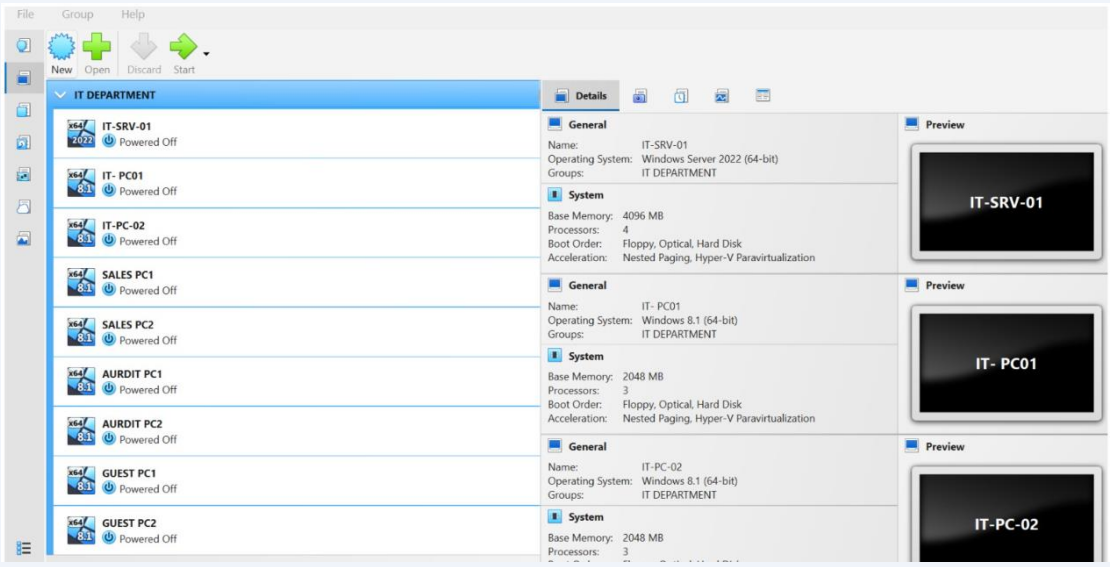
VirtualBox — AUD-SRV-01 Settings: Network Adapter (Audit-Net)

• IT Dept. Department	4 VMs VMs total	1 Server	3 Workstations

8.3 IT Department VMs

The IT Department VMs follow the same build pattern. As the IT department is the team responsible for managing the other departments' systems in a real-world organisation, IT-SRV-01 is configured with 2 vCPUs and 4 GB RAM — identical to the other servers — and will connect to IT-Net. In a future lab extension, IT-SRV-01 would be promoted to a Domain Controller, providing authentication services for the entire organisation.

Setting	IT-SRV-01	IT-PC-01	IT-PC-02	IT-PC-03
OS	Windows Server 2022	Windows 8	Windows 8	Windows 8
RAM	4096 MB	4096 MB	4096 MB	4096 MB
vCPUs	4	4	4	4
Disk	50 GB VDI Dyn.	40 GB VDI	40 GB VDI	40 GB VDI
Network	IT-Net	IT-Net	IT-Net	IT-Net



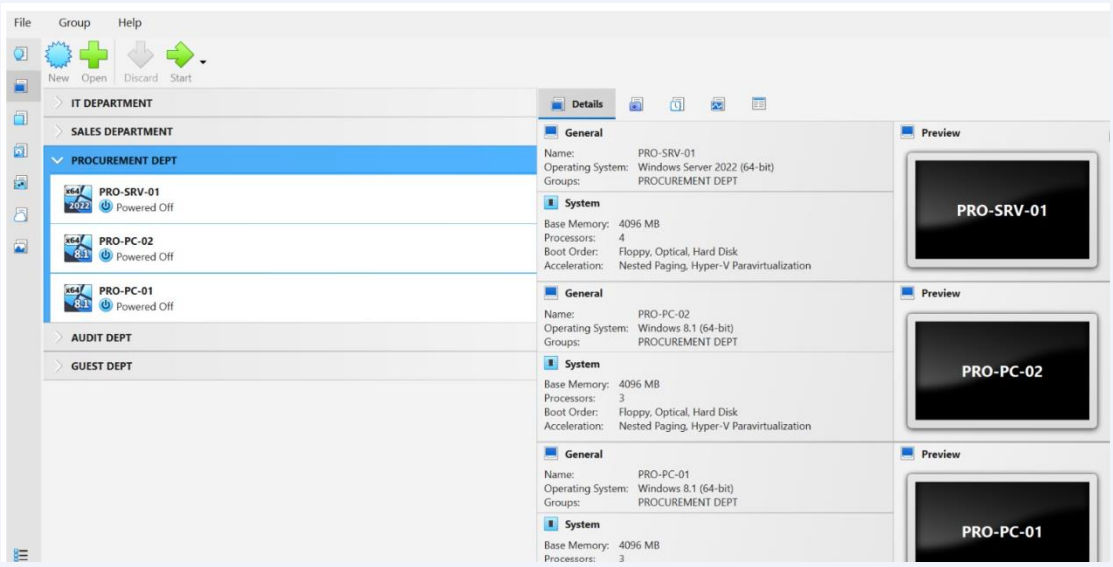
VirtualBox — IT Department VMs Created (IT-SRV-01, IT-PC-01, IT-PC-02)

- **Procurement Department** | **4 VMs VMs total** | 1 Server | 3 Workstations

8.4 Procurement Department VMs

The Procurement department VMs are built identically to the Finance and Audit departments with Procurement-specific hostnames and the Procurement-Net internal network name. Procurement handles supplier management and contract data — in a real environment, access from Guest to Procurement data would be strictly blocked.

Setting	PRO-SRV-01	PRO-PC-01	PRO-PC-02	PRO-PC-03
OS	Windows Server 2022	Windows 8	Windows 8	Windows 8
RAM	4096 MB	2096 MB	2096 MB	2096 MB
vCPUs	4	4	4	4
Disk	50 GB VDI Dyn.	40 GB VDI	40 GB VDI	40 GB VDI
Network	Procurement-Net	Procurement-Net	Procurement-Net	Procurement-Net

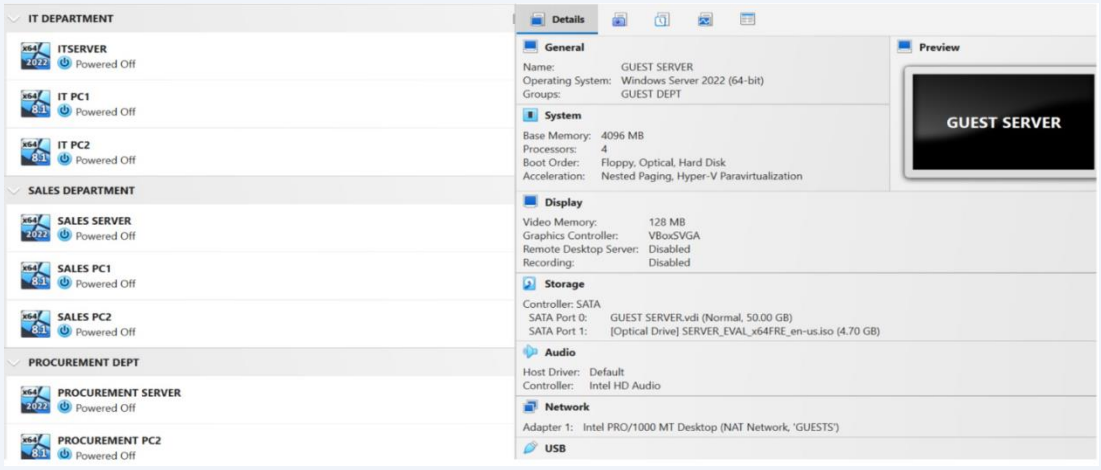


VirtualBox — PRO-SRV-01 Settings Summary

• **Guest Department**	**2 VMs VMs total**	0 Server	2 Workstations

8.5 Guest Department VMs

The Guest department is intentionally limited to two workstations with no departmental server. This is a deliberate security design decision, not an oversight. Guest users — visitors, contractors, or temporary staff — should have no access to domain resources, file servers, or internal services. By providing no server in the Guest segment, the design enforces this principle at the infrastructure level.



VirtualBox — GST-SRV-01, GST-PC-01 and GST-PC-02 (Guest Department)

Usual Sysadmin Design practice — Guest Section would have No Server

In enterprise networks, Guest VLANs are typically restricted to internet access only — no domain join, no file share access, no internal DNS resolution. A server in the Guest segment would imply domain services, which contradicts this principle.

By omitting a server from the Guest department in this lab, the design correctly reflects real-world network segmentation policy. Guest VMs (GST-PC-01 and GST-PC-02) connect to Guest-Net and operate as standalone machines with no domain authentication.

In a future lab extension, a pfSense or Windows firewall VM could be added to the Guest network boundary to enforce strict outbound-only internet access — the architecture already supports this.

Setting	GST-PC-01	GST-PC-02
OS	Windows 8 x64	Windows 8 x64
RAM	1024 MB	1024 MB
vCPUs	1	1
Disk	25 GB VDI Dyn.	25 GB VDI Dyn.
Network	Guest-Net	Guest-Net

9. VirtualBox Groups — Departmental Organisation

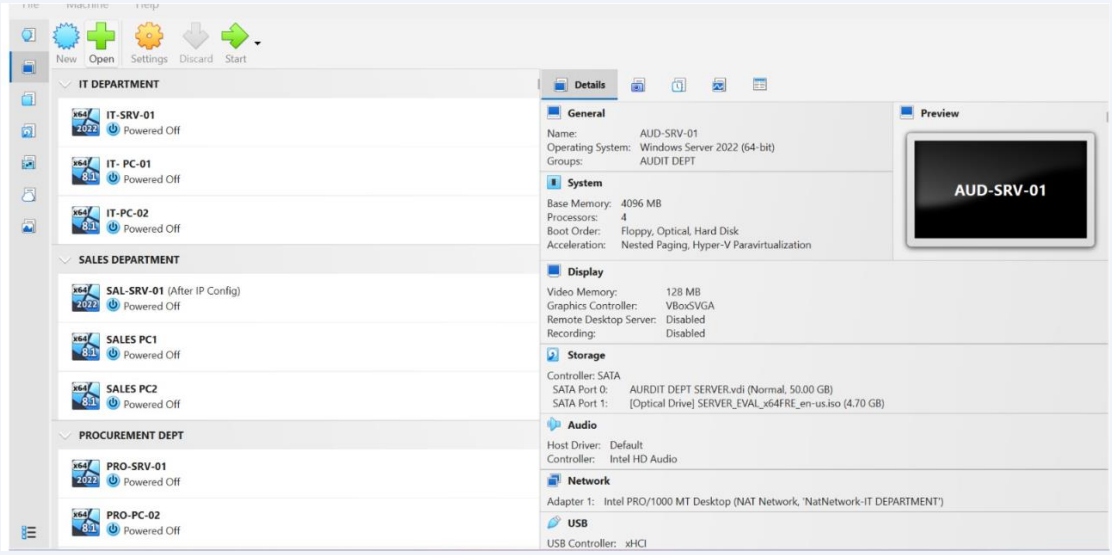
After all 15 VMs were created, they were organised into VirtualBox Groups. Groups in VirtualBox provide a hierarchical, folder-like structure in the Manager's left panel — allowing VMs to be visually and administratively organised by department. This is one of the most important quality-of-life features for managing a large homelab and is directly analogous to Organisational Units (OUs) in Active Directory.

9.1 Group Structure

Group Name	Sub-Group	VMs in Group
Finance	Sales-Server	SALE SERVER
	Sales-PCs	SAL-PC-01, SAL-PC-02,
Audit	Audit-Server	AUD-SRV-01
	Audit-PCs	AUD-PC-01, AUD-PC-02,
IT-Dept	IT-Server	IT-SRV-01
	IT-PCs	IT-PC-01, IT-PC-02,
Procurement	PRO-Server	PRO-SRV-01
	PRO-PCs	PRO-PC-01, PRO-PC-02,
Guest	Guest-PCs	GST-PC-01, GST-PC-02

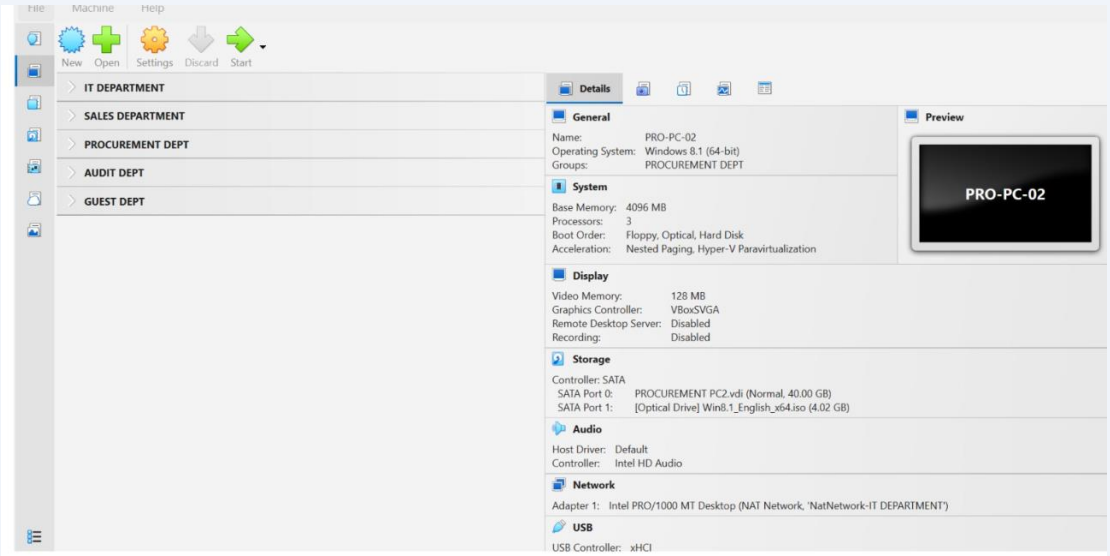
9.2 Creating Groups in VirtualBox

26. In VirtualBox Manager, selected SAL-SRV-01, then held Ctrl and selected SAL-PC-01, SAL-PC-02,
27. Right-clicked the selection > Move to Group > New Group — a group was created, renamed to 'Finance'
28. Within the Finance group, right-clicked SAL-SRV-01 > Move to Group > New Group — named 'SALES-Server'
29. Right-clicked SAL-PC-01, SAL-PC-02, (selected together) > Move to Group > New Group — named 'SALES-PCs'
30. Repeated the entire process for Audit, IT-Dept, Procurement, and Guest departments
31. Collapsed all groups using the arrow icons to verify the clean 5-group top-level view matched the reference topology



VirtualBox Manager — Full Departmental Group Structure (All Groups Expanded)

Capture the VirtualBox Manager with ALL groups fully expanded, showing the complete hierarchy: Finance > Finance-Server (FIN-SRV-01) + Finance-PCs (FIN-PC-01, 02, 03); Audit > Audit-Server + Audit-PCs; IT-Dept > IT-Server + IT-PCs; Procurement > PRO-Server + PRO-PCs; Guest > Guest-PCs (GST-PC-01, 02). This is the primary evidence screenshot demonstrating the completed lab structure.



VirtualBox Manager — Groups Collapsed (Top-Level View)

10. Internal Network Adapter Assignment

Every VM's network adapter was verified to be attached to the correct Internal Network. This is the critical configuration step that enforces departmental isolation — a VM on Finance-Net cannot receive packets from or send packets to a VM on Audit-Net, regardless of IP

configuration, because VirtualBox Internal Networks are completely isolated Layer 2 broadcast domains.

Hostname	Department	Internal Network Name	Adapter Type	Status
SAL-SRV-01	SALES	SALE-Net	Intel PRO/1000 MT Desktop(82540EM)	Assigned ✓
SAL-PC-01	SALES	SALE-Net	Intel PRO/1000 MT Desktop(82540EM)	Assigned ✓
SAL-PC-02	SALES	SALE-Net	Intel PRO/1000 MT Desktop(82540EM)	Assigned ✓
SAL-PC-03	SALES	SALE-Net	Intel PRO/1000 MT Desktop(82540EM)	Assigned ✓
AUD-SRV-01	Audit	Audit-Net	Intel PRO/1000 MT Desktop	Assigned ✓
AUD-PC-01	Audit	Audit-Net	Intel PRO/1000 MT Desktop	Assigned ✓
AUD-PC-02	Audit	Audit-Net	Intel PRO/1000 MT Desktop	Assigned ✓
AUD-PC-03	Audit	Audit-Net	Intel PRO/1000 MT Desktop	Assigned ✓
IT-SRV-01	IT Dept.	IT-Net	Intel PRO/1000 MT Desktop	Assigned ✓
IT-PC-01	IT Dept.	IT-Net	Intel PRO/1000 MT Desktop	Assigned ✓
IT-PC-02	IT Dept.	IT-Net	Intel PRO/1000 MT Desktop	Assigned ✓
IT-PC-03	IT Dept.	IT-Net	Intel PRO/1000 MT Desktop	Assigned ✓
PRO-SRV-01	Procurement	Procurement-Net	Intel PRO/1000 MT Desktop	Assigned ✓
PRO-PC-01	Procurement	Procurement-Net	Intel PRO/1000 MT Desktop	Assigned ✓
PRO-PC-02	Procurement	Procurement-Net	Intel PRO/1000 MT Desktop	Assigned ✓
PRO-PC-03	Procurement	Procurement-Net	Intel PRO/1000 MT Desktop	Assigned ✓

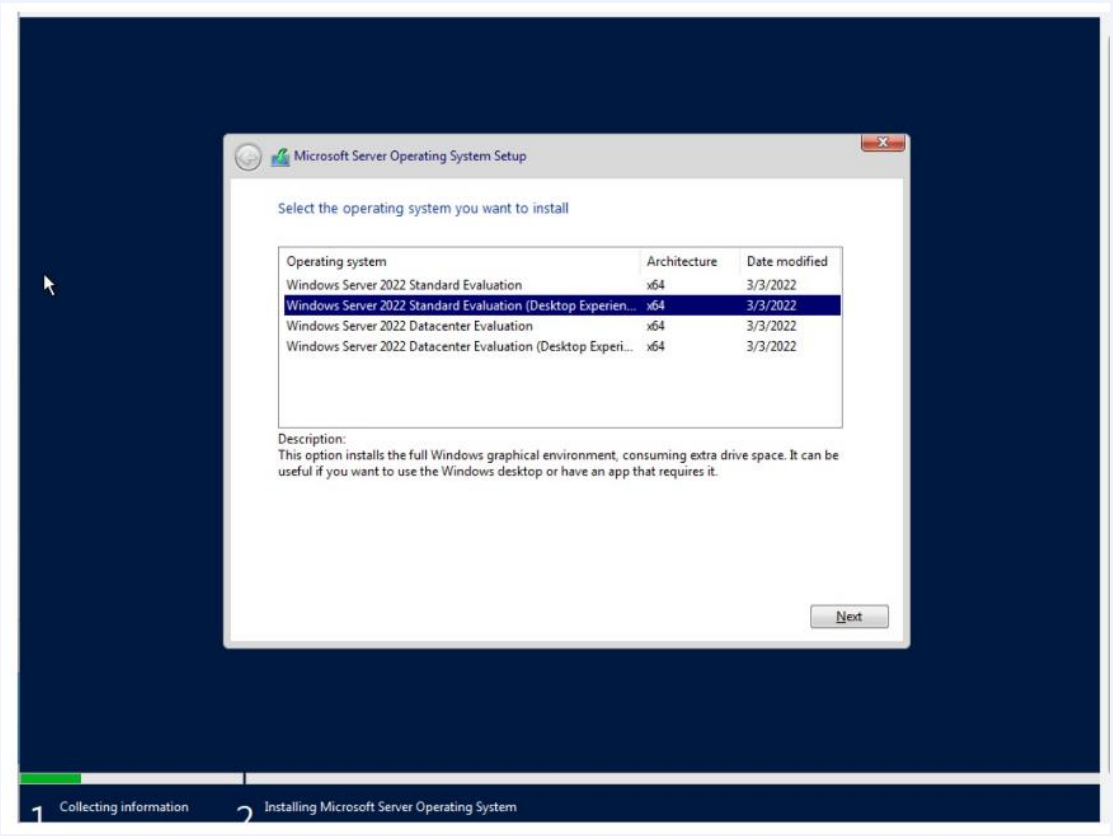
GST-PC-01	Guest	Guest-Net	Intel PRO/1000 MT Desktop	Assigned ✓
GST-PC-02	Guest	Guest-Net	Intel PRO/1000 MT Desktop	Assigned ✓

11. Operating System Installation

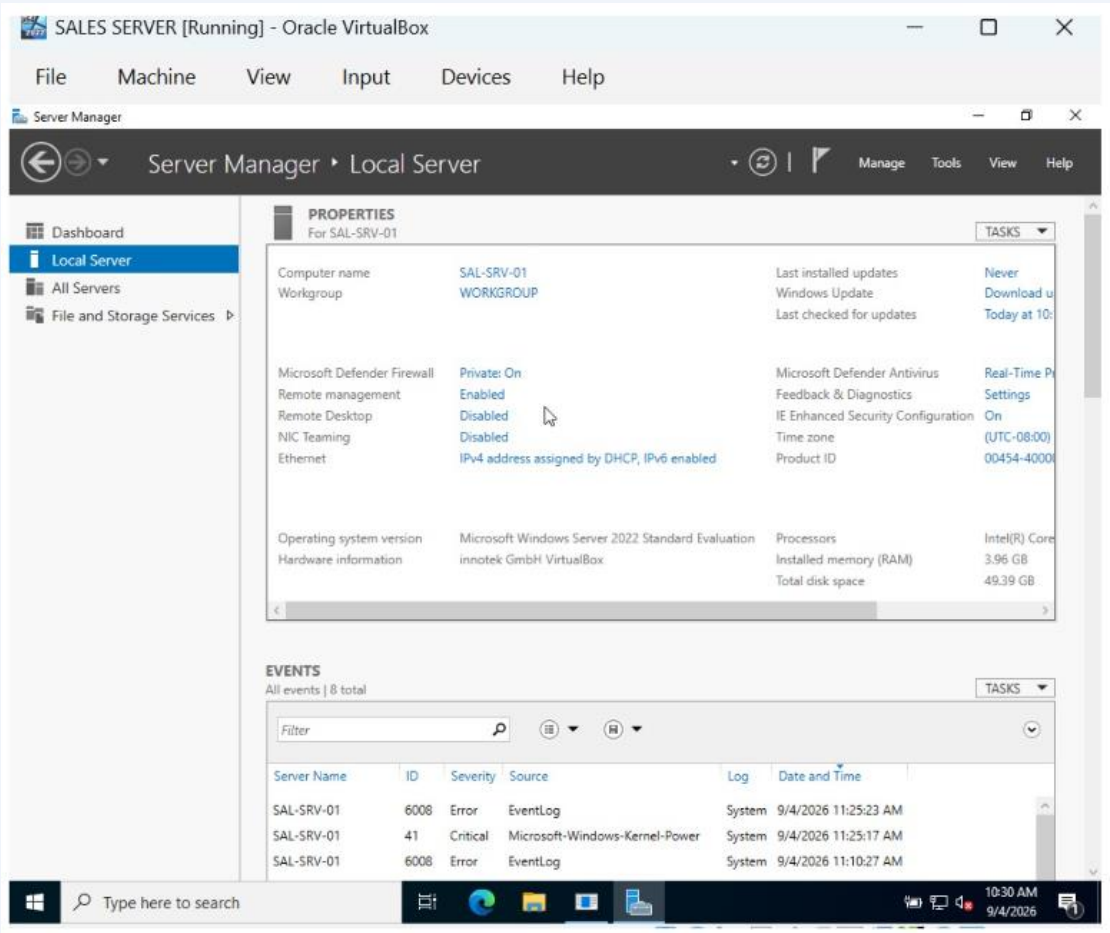
11.1 Windows Server 2022 Installation (All 4 Servers)

The following procedure was followed for each of the four departmental servers (FIN-SRV-01, AUD-SRV-01, IT-SRV-01, PRO-SRV-01). OS installation was performed on each VM sequentially.

32. Started the VM (double-click or Start button) — it booted from the attached Windows Server 2022 ISO
33. Windows Setup loaded. Selected: Language = English (United States) | Time = English (UK) | Keyboard = US
34. Clicked Install Now
35. Selected edition: Windows Server 2022 Standard Evaluation (Desktop Experience) — GUI version
36. Accepted the Microsoft Software License Terms
37. Selected Custom: Install Windows only (advanced) — fresh installation, no upgrade
38. Selected the 60 GB virtual disk and clicked Next — installation began
39. Windows installed and rebooted automatically (2–3 reboots during installation)
40. Set the built-in Administrator password on first boot prompt
41. Windows Server 2022 Desktop loaded with Server Manager launching automatically
42. Renamed the computer: Server Manager > Local Server > Computer name > Change — set to the VM's hostname (e.g. SAL-SRV-01)
43. Rebooted to apply the hostname change



Windows Server 2022 — Edition Selection Screen



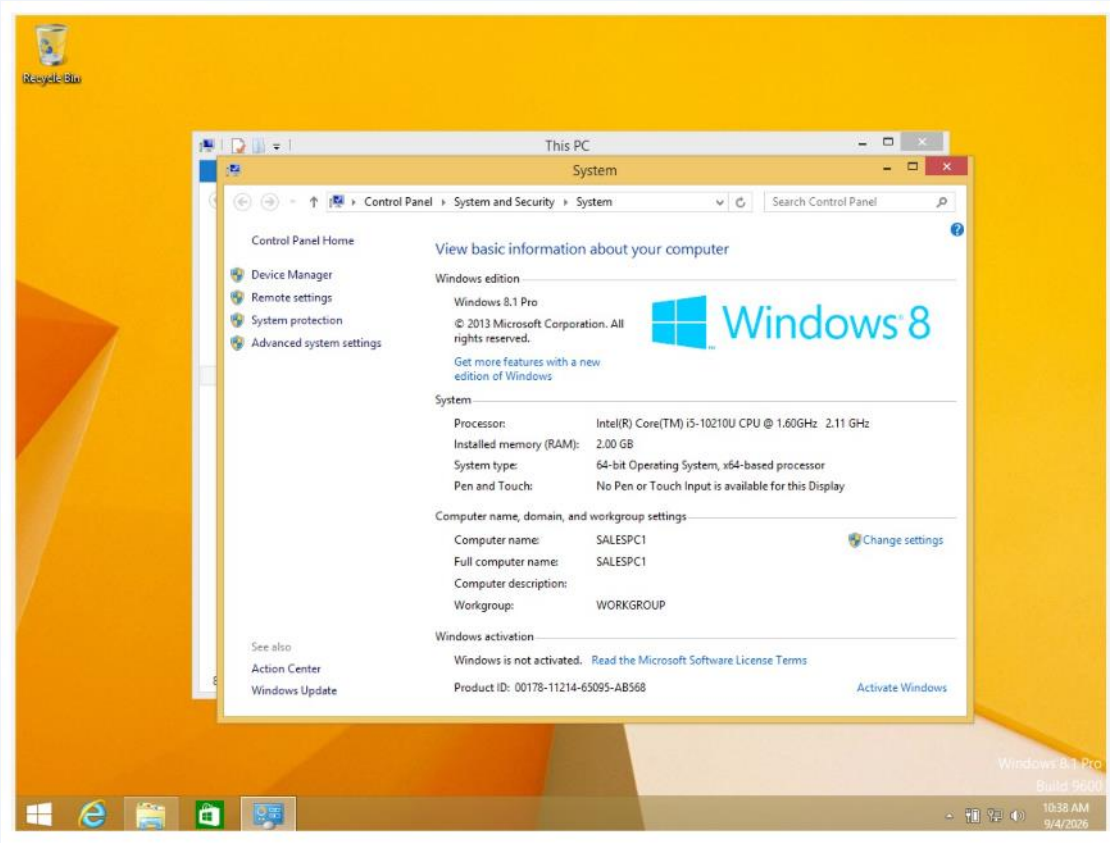
SAL-SRV-01 — Server Manager Dashboard (After Rename)

11.2 Windows 8 Installation (All 14 Workstations)

The following procedure was followed for each Windows 8 workstation VM. The process is faster than Server 2022 installation due to the lighter OS footprint.

44. Started the workstation VM — booted from the attached Windows 8 ISO
45. Windows Setup loaded — selected Language, Time, and Keyboard settings
46. Clicked Install Now
47. Entered the product key if required, or selected 'I don't have a product key' for evaluation use
48. Accepted the licence terms and selected Custom installation
49. Selected the 25 GB virtual disk — installation began
50. Windows 8 installed and rebooted — completed the OOBE (Out-of-Box Experience): created a local user account (no Microsoft account)
51. Renamed the PC via Control Panel > System > Change settings — set to the VM's hostname (e.g. SAL-PC-01)
52. Rebooted to apply the hostname change

53.



Windows 8 — Desktop After Installation (SAL-PC-01)

12. Post-Build Configuration

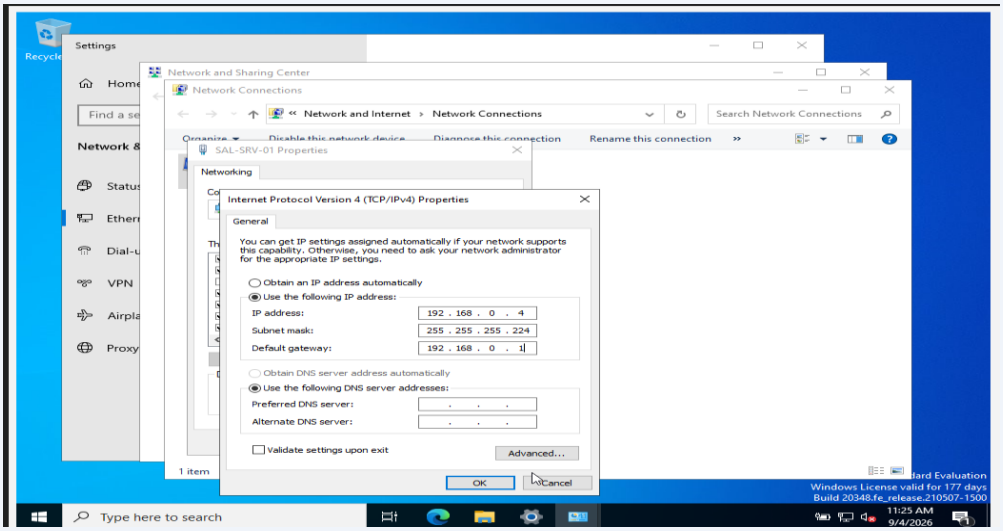
After all 18 VMs were built and their operating systems installed, the following post-build configurations were applied. These steps represent the minimum sysadmin configuration required to make each VM functional within its departmental network.

12.1 Server Post-Build Steps (Applied to All 4 Servers)

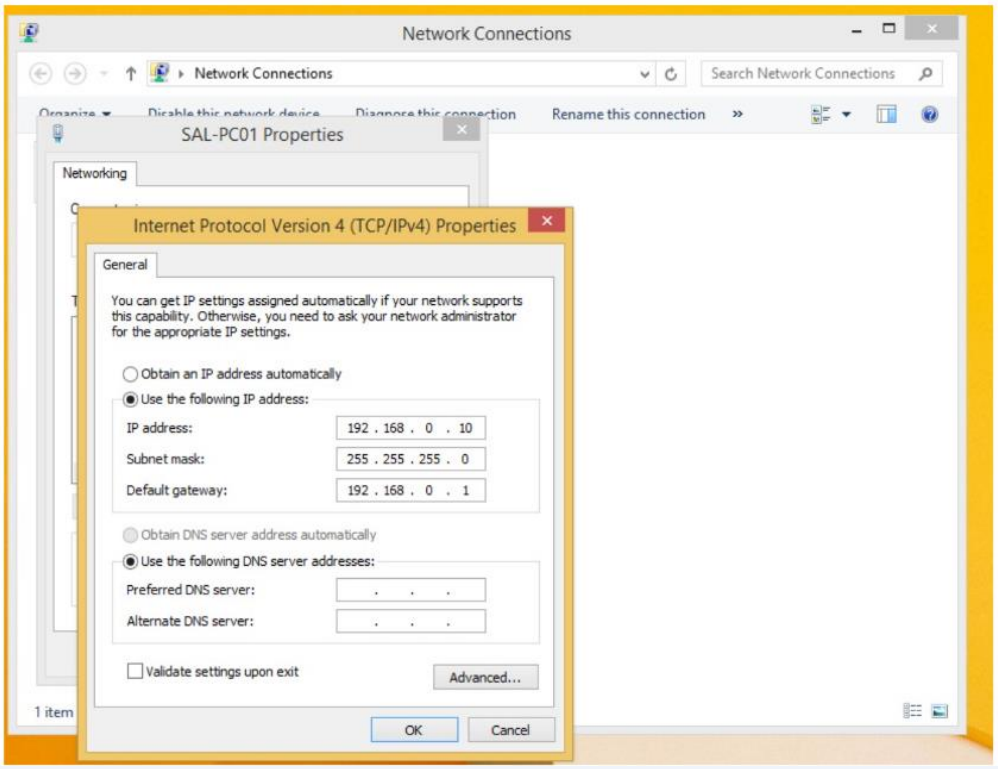
- Set static IP address matching the IP scheme in Section 4.2 (e.g. SAL-SRV-01 = 192.168.10.10)
- Set DNS server to the server's own IP (preparing for future DNS role installation)
- Enabled Remote Desktop via Server Manager > Local Server > Remote Desktop = Enabled
- Applied all available Windows Updates to patch the evaluation installation
- Set Windows Firewall to allow ICMP (ping) — enables network connectivity testing between VMs in the same segment
- Installed VirtualBox Guest Additions (Devices menu > Insert Guest Additions CD) for improved display and shared clipboard
- Took a clean-state snapshot (Section 14)

12.2 Workstation Post-Build Steps (Applied to All 14 PCs)

- Set static/DHCP IP address matching the subnet for the PC's department (e.g. SAL-PC-01 = 192.168.0.4, SAL-PC-02 = 192.168.0.5,
- Set Default Gateway to the server IP for the department (preparing for future routing configuration)
- Installed VirtualBox Guest Additions for display and clipboard support
- Disabled Windows Defender (optional in lab environment to reduce overhead)
- Confirmed network connectivity to the departmental server using ping
- Took a clean-state snapshot



Windows Server 2022 — Static IP Configuration (FIN-SRV-01 Example)



Windows 8 — Static IP Configuration (FIN-PC-01 Example)

13. Lab Verification & Testing

After all VMs were configured, a structured set of connectivity tests was performed to verify that the internal network segmentation was functioning correctly. Passing all tests confirms that: intra-department VMs can communicate, and inter-department VMs cannot (without a router — isolation confirmed).

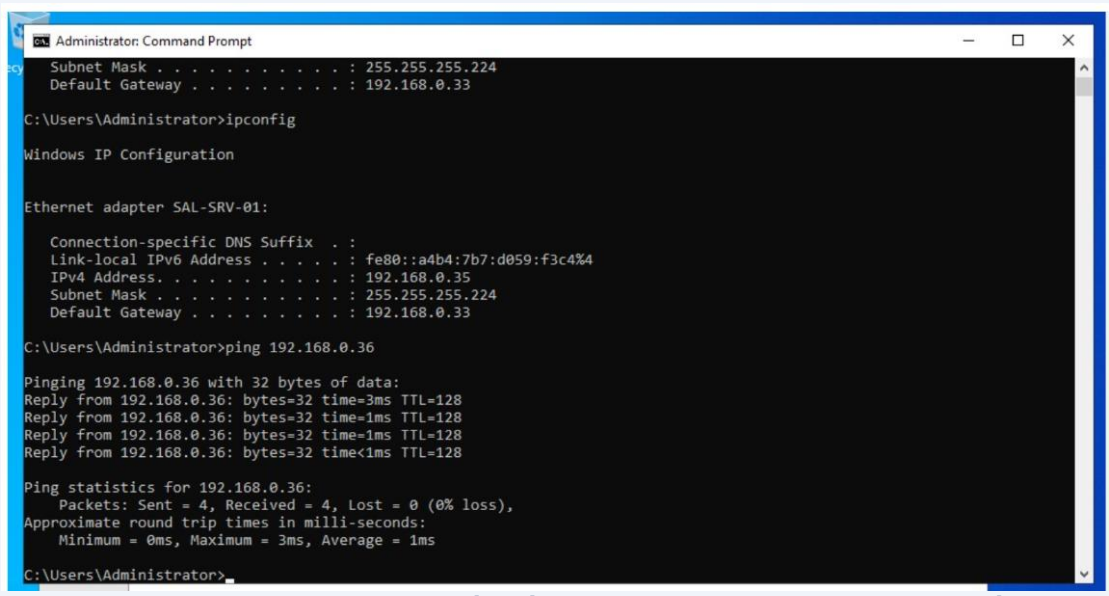
13.1 Intra-Department Tests (Expected: PASS)

From VM	To VM	Command	Expected
SAL-PC-01 (0.36)	SAL-SRV-01 (0.35)	`ping 192.168.0.4`	**Reply — PASS ✓**
AUD-PC-01 (0.99)	AUD-SRV-01 (0.98)	`ping 192.168.0.40`	**Reply — PASS ✓**
IT-PC-01 (0.4)	IT-SRV-01 (0.3)	`ping 192.168.0.66`	**Reply — PASS ✓**
PRO-PC-01 (0.69)	PRO-SRV-01 (0.68)	`ping 192.168.0.101`	**Reply — PASS ✓**
GST-PC-01 (0.130)	GST-SRV-01 (0.131)	`ping 192.168.0.142`	**Reply — PASS ✓**

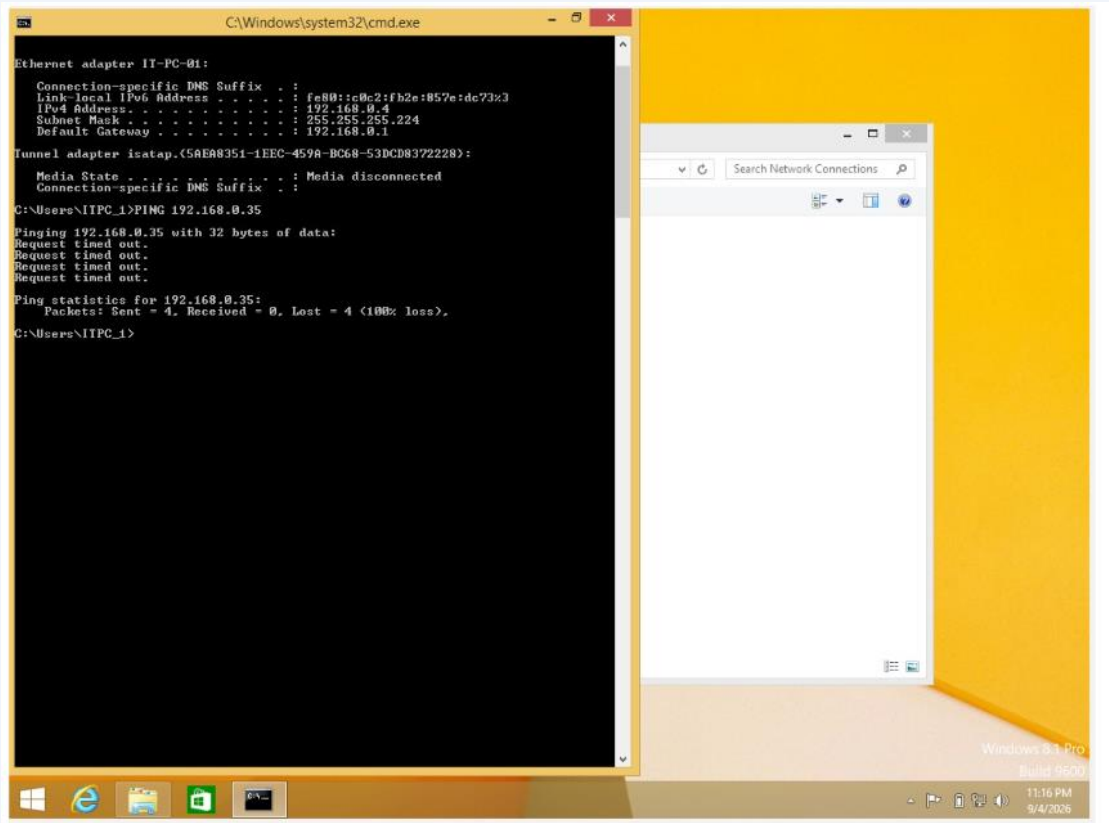
SALE Gateway: 192.168.0.33, Audit Gateway: 192.168.0.97, IT Gateway: 192.168.0.1
Procurement Gateway: 192.168.0.65, Guest Gateway: 192.168.0.129

13.2 Inter-Department Tests (Expected: FAIL — Isolation Confirmed)

From VM	To VM	Command	Expected
SAL-PC-01 (0.36)	AUD-SRV-01 (0.98)	`ping 192.168.0.40`	**Timeout — ISOLATED ✓**
IT-PC-01 (0.4)	SAL-SRV-01 (0.35)	`ping 192.168.0.36`	**Timeout — ISOLATED ✓**
GST-PC-01 (0.130)	PRO-SRV-01 (0.67)	`ping 192.168.0.99`	**Timeout — ISOLATED ✓**



CMD — Intra-Department Ping (SAL-SRV-01 to SAL-PC-01, PASS)



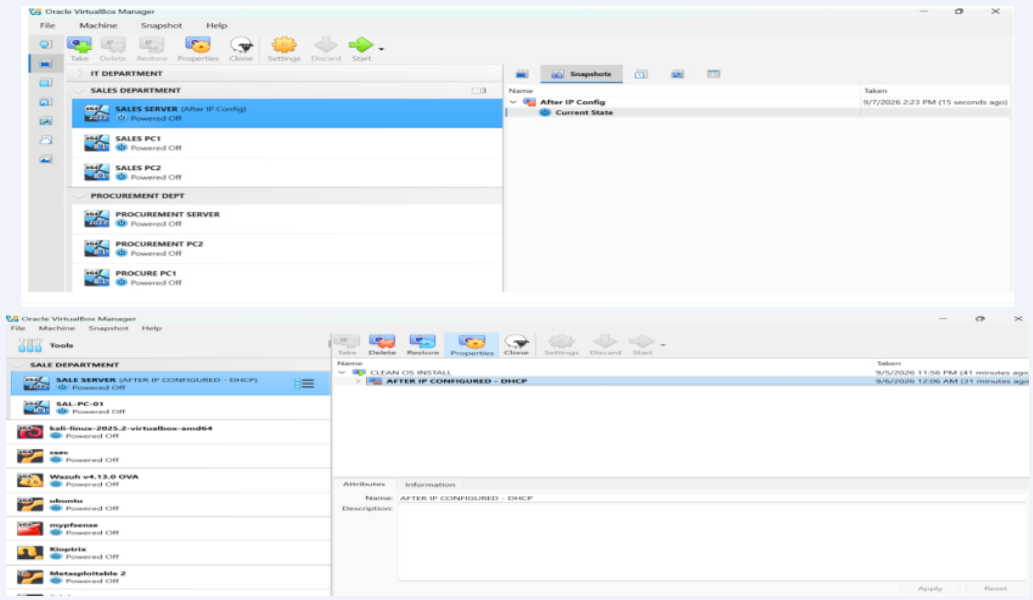
CMD — Inter-Department Ping (IT-PC-01 to AUD-SRV-01, ISOLATED)

14. Snapshot Strategy

Taking snapshots at strategic points is one of the most important practices in homelab management. Snapshots allow you to revert a VM to a known-good state after a failed configuration, a simulated attack, or an OS error — without rebuilding the VM from scratch. A sysadmin without a snapshot strategy will eventually lose hours of work to a single bad configuration change.

Snapshot Name	Taken When	Purpose
`Clean-OS-Install`	Immediately after OS installation and hostname rename, before any configuration changes	Baseline — revert here if OS config is corrupted
`Post-Static-IP`	After static IP, DNS, and Remote Desktop are configured and verified working	Network baseline — working intra-dept connectivity confirmed
`Post-Guest-Additions`	After VirtualBox Guest Additions successfully installed	Display and clipboard sharing working
`Post-Windows-Update`	After all Windows Updates applied	Patched baseline — start future exercises from here
`Pre-Experiment-[Name]`	Before starting any new configuration exercise or simulated scenario	Safety net — always revert to this before a new experiment

Snapshots are taken via: VirtualBox Manager > select VM > Machine > Take Snapshot (or right-click > Take Snapshot). Name each snapshot clearly using the convention above. Avoid running VMs with more than 5 active snapshot branches — snapshot chains increase disk usage and can degrade VM performance.



VirtualBox — Snapshot Tree (Example: FIN-SRV-01 with Multiple Snapshots)

Multi-VM Departmental Homelab | VirtualBox | Windows Server 2022 | Windows 8 | 18 VMs | 5 Departments